

# Skills Summary

## Completing Two-Column Triangle-Congruence Proofs

*Use this reference while completing your worksheet or when studying.*

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### Skill 1 — Supplying the reason for a line

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Every reason is one of a short list: **Given**; a **definition** (midpoint, angle bisector, right triangle, congruent segments); the **reflexive property** for a shared side or angle; a **theorem** (vertical angles congruent, alternate interior angles congruent); or a **criterion** — SSS, SAS, ASA, AAS, HL — on the final line.

**Two reasons no mark ever shows:** the reflexive property for a shared part, and vertical angles at a crossing. Look for them before deciding a proof is short of evidence.

**Example:** A proof reaches  $\triangle ABC \cong \triangle CDA$  by SSS using a shared diagonal, so by CPCTC two corresponding sides are equal:  $2x + 9$  and  $4x - 5$  cm. Find  $x$ .

**Step 1.** The shared diagonal enters by the reflexive property, which completes SSS; only then does CPCTC let me set the two corresponding lengths equal.

**Step 2.**  $2x + 9 = 4x - 5$ , so  $14 = 2x$ .

$$\Rightarrow x = 7$$

**Try it:** After a congruence proof, CPCTC gives two corresponding sides  $6x - 1$  and  $3x + 20$  cm. Find  $x$ .

check:  $7 = x$

### Skill 2 — Supplying the statement for a reason

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Read the reason column and work backwards. “Definition of midpoint” must be followed by the congruence of the two halves; “reflexive property” must be the shared part; the criterion on the last line tells you what the lines above it have to contain.

**The criterion names the pattern:** ASA needs angle, included side, angle; SAS needs side, included angle, side; AAS needs angle, angle, non-included side. If your statements do not match that pattern, one of them is in the wrong place.

**Free third angle:** two matching angles force the third, since the angles of a triangle sum to  $180^\circ$ .

**Example:** Two angles of a triangle measure  $a = 63^\circ$  and  $b = 49^\circ$ . Find the third angle, the statement an “angle sum” reason would carry.

**Step 1.** The reason cited is the triangle angle sum, so the statement it justifies must be the value of the remaining angle.

**Step 2.** Third angle  $= 180 - a - b = 180 - 63 - 49$ .

$$\Rightarrow 68 \text{ degrees}$$

**Try it:** Two angles of a triangle measure  $55^\circ$  and  $82^\circ$ . Find the third.

check:  $43$

### Skill 3 — Finishing with CPCTC

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**CPCTC** — corresponding parts of congruent triangles are congruent — is the step *after* the criterion. It converts a proved triangle congruence into a congruence of any one pair of remaining sides or angles.

**Order matters:** you may never use CPCTC to establish the criterion. The proof runs  
givens  $\rightarrow$  criterion  $\rightarrow$  CPCTC  $\rightarrow$  equation.

Then set the two corresponding expressions equal and solve, and substitute back into both to check.

**Example:** Two triangles are proved congruent, and CPCTC gives corresponding angles of  $(7x - 12)^\circ$  and  $(4x + 15)^\circ$ . Find  $x$ .

**Step 1.** CPCTC comes after the criterion, so the congruence is already established and the two corresponding measures may simply be set equal.

**Step 2.**  $7x - 12 = 4x + 15$ , so  $3x = 27$ .

$$\Rightarrow x = 9$$

**Try it:** CPCTC gives corresponding angles of  $(5x + 6)^\circ$  and  $(2x + 33)^\circ$ . Find  $x$ .

**6 = x** :check